

# Customer Lifetime Value Prediction Model

Project Report | Online Retail Customer Analytics

## Abstract

This project estimates future customer value from historical purchase behavior in the Online Retail dataset. Transaction records were cleaned, aggregated into customer-level features, and used to train a Random Forest Regression model. The final workflow produces customer-level predictions and a reusable saved model, supporting practical customer value analysis.

## Introduction

Customer Lifetime Value (LTV) helps organizations understand which customers are likely to contribute the most future value. This project applies machine learning to retail transactions to predict future customer value from each customer's previous purchasing behavior. The analysis uses the Online Retail dataset, which contains 541,909 transactions across 8 columns.

## Tools Used

|                   |                                                      |
|-------------------|------------------------------------------------------|
| Python            | Data preparation and model workflow                  |
| Pandas and NumPy  | Data cleaning and customer-level feature engineering |
| Scikit-learn      | Random Forest Regression and model evaluation        |
| Matplotlib        | Project visualizations                               |
| Excel and VS Code | Dataset handling and development environment         |

## Steps Involved

- 1. Data cleaning:** Removed records without Customer IDs, cancelled orders, and invalid quantities or prices. This resulted in 397,884 valid transactions.
- 2. Customer feature engineering:** Calculated Recency, Frequency, TotalSpend, and Average Order Value (AOV) from historical transactions.
- 3. Target creation:** Used a cutoff date of 2011-09-01 to define future customer value. The resulting model dataset contained 3,317 customers.
- 4. Model development:** Split the customer-level data into training and testing sets, then trained a Random Forest Regression model.
- 5. Evaluation and outputs:** Evaluated predictions using MAE and RMSE, produced customer LTV predictions, and saved the trained model.

## Results and Model Performance

| Metric                         | Result  |
|--------------------------------|---------|
| Mean Absolute Error (MAE)      | 944.64  |
| Root Mean Squared Error (RMSE) | 4651.88 |

The model's test-set performance is summarized above. MAE measures the average absolute difference between predicted and actual future customer value, while RMSE places greater weight on larger errors. These metrics are reported directly from the project evaluation.

## Deliverables and Visualizations

|                              |                                                      |
|------------------------------|------------------------------------------------------|
| Customer_LTV_Predictions.csv | Customer-level future value predictions              |
| customer_ltv_model.pkl       | Saved Random Forest Regression model                 |
| ltv_distribution.png         | Distribution of customer future value                |
| actual_vs_predicted_ltv.png  | Comparison of actual and predicted customer LTV      |
| feature_importance.png       | Relative importance of the engineered model features |

## Conclusion

This project demonstrates an end-to-end customer LTV prediction workflow: cleaning retail transactions, converting purchase history into meaningful customer features, defining future value with a time-based cutoff, and training a Random Forest Regression model. The resulting prediction file, trained model, and visualizations provide a complete foundation for customer-level value analysis.

**Key project facts: 541,909 raw transactions | 397,884 valid transactions after cleaning | 4,338 customers overall | 3,317 customers in the model dataset**